08/10/2025

Code-A Phase-1

Aakash
Medical | IIT-JEE | Foundations

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MM : 720

NCERT Booster Test Series for NEET-2026_RM(P1)_NBTS-02A

Time : 180 Min.

Topics Covered:
Physics: Gravitation, Mechanical Properties of Solids, Mechanical Properties of Fluids, Thermal Properties of Matter, Thermodynamics, Kinetic Theory, Oscillations, Waves, Electric Charges & Fields
Chemistry: Thermodynamics, Equilibrium, Redox Reactions, Electrochemistry, Solutions
Botany: Morphology of Flowering Plants, Anatomy of Flowering Plants, Plant Kingdom
Zoology: Excretory Products & their Elimination, Locomotion & Movement, Neural Control & Coordination, Chemical Coordination & Integration

General Instructions :
Duration of Test is 3 hrs.
The Test consists of 180 questions. The maximum marks are 720.
There are four parts in the question paper consisting of Physics, Chemistry, Botany and Zoology having 45 questions in each part of equal weightage.
Each question carries +4 marks. For every wrong response, –1 mark shall be deducted from the total score. Unanswered/unattempted questions will be given no marks.
Use blue/black ballpoint pen only to darken the appropriate circle.
Mark should be dark and completely fill the circle.
Dark only one circle for each entry.
Dark the circle in the space provided only.
Rough work must not be done on the Answer sheet and do not use white fluid or any other rubbing material on the Answer sheet.

PHYSICS

1. Which of the following is/are Kepler's laws?
   (1) Law of orbit
   (2) Law of areas
   (3) Law of continuity
   (4) Both (1) and (2)

2. Which of the following is correct dimensional formula for gravitational field intensity?
   (1) $[MLT^{-2}]$
   (2) $[L^2T^{-2}]$
   (3) $[M^{-1}L^1T^{-2}]$
   (4) $[LT^{-2}]$

3. Choose the correct statement.
   (1) Gravitational potential as well as gravitational field inside the shell are not zero
   (2) Only gravitational field inside a shell is zero
   (3) Gravitational field and gravitational potential both are zero inside a shell
   (4) Only gravitational potential inside the shell is zero

4. Acceleration due to gravity on the surface of earth is g. In column-I different values of acceleration due to gravity is given and in column-II different values of height and depth are given.
Match the columns and tick the **correct** option.
(R is radius of earth)

	Column-I		Column-II
a	$\frac{g}{2}$	(i)	Depth = R
b.	$\frac{g}{4}$	(ii)	Depth = $\frac{R}{2}$
c.	$\frac{g}{9}$	(iii)	Height = $2R$
d.	Zero	(iv)	Height = R

(1) a(ii), b(iii), c(iv), d(i)

(2) a(ii), b(iv), c(iii), d(i)

(3) a(i), b(ii), c-(iv), d(iii)

(4) a(iv), b(iii), c(i), d(ii)

5. When a planet comes closer to sun in an elliptical orbit, its linear speed

(1) Increases

(2) Decreases

(3) Remains same

(4) May be (1) or (2) depending on mass of planet

6. **Assertion (A) :** A satellite moving in a circular orbit around the earth has a total energy $-E_0$, then its kinetic energy is E_0.
Reason (R) : Potential energy is equal to negative of total energy of the satellite moving in a circular orbit.

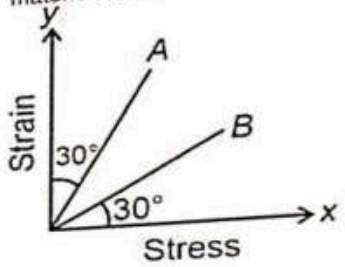
(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

7. The strain-stress graph for wires A and B are as shown in the figure. If Y_A and Y_B are the Young's moduli of the materials, then



(1) $2Y_A = Y_B$

(2) $Y_A = Y_B$

(3) $Y_A = 3Y_B$

(4) $3Y_A = Y_B$

8. The Bulk modulus of a spherical object is B. If it is subjected to a uniform pressure P from all sides, the fractional decrease in its area will be

(1) $\frac{P}{B}$

(2) $\frac{P}{2B}$

(3) $\frac{P}{3B}$

(4) $\frac{2}{3}\frac{P}{B}$

9. A uniform rod of length L has mass M. If the area of cross-section is A and Young's modulus of rod is Y, then elongation in rod due to its own weight will be

(1) $\frac{MgL}{AY}$

(2) $\frac{MgL}{2AY}$

(3) $\frac{MgL}{2AY}$

(4) $\frac{2MgL}{AY}$

10. If the length of wire is increased by 0.02% and the young's modulus for the wire is $Y = 2 \times 10^9$ N/m^2, then the potential energy density stored in the wire is

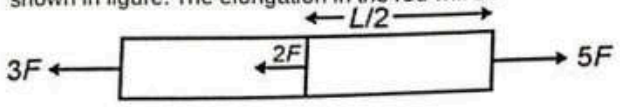
(1) 40 J/m^3

(2) 80 J/m^3

(3) 160 J/m^3

(4) 60 J/m^3

11. A uniform cylindrical rod of length L, cross-sectional area A and Young's modulus Y is acted upon by the forces as shown in figure. The elongation in the rod will be



(1) $\frac{2FL}{5AY}$

(2) $\frac{3FL}{8AY}$

(3) $\frac{4FL}{AY}$

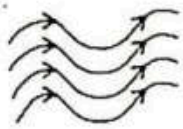
(4) $\frac{8FL}{3AY}$

12. Which of the following options represents uniform electric field?

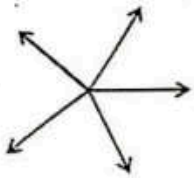
(1)



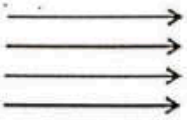
(2)



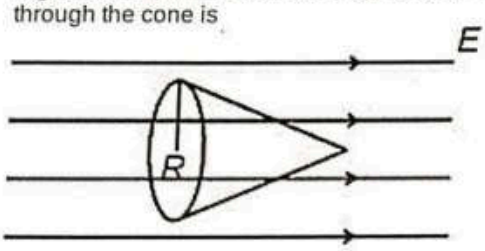
(3)



(4)



13. A closed Gaussian surface in form of a cone is formed in a region of uniform electric field as shown. The electric flux through the cone is



(1) $\pi R^2 E$

(2) $2\pi RE$

(3) $2\pi R^2 E$

(4) Zero

14. Charge is uniformly distributed with volume charge density 2ρ in a spherical volume of radius R. A cavity of radius r is made in charge distribution such that the centre of cavity is at distance a from the centre of the charge distribution then the electric field in the cavity is

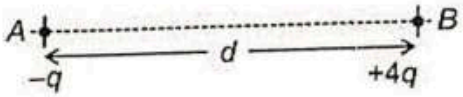
(1) Zero

(2) Non zero and non–uniform

(3) Non zero, uniform and equal to $\frac{\rho}{3\varepsilon_0}\vec{a}$

(4) Non zero, uniform and equal to $\frac{2\rho}{3\varepsilon_0}\vec{a}$

15. For two point charges placed at distance d as shown below, the electric field will be zero at



(1) Mid way between A and B

(2) Distance d from $(-q)$ and to the left

(3) Distance d from $(4q)$ to the right

(4) No finite distance

16. Two particles having charges $+Q$ and $-Q$ are placed at certain distance apart and force acting between them is F. If 50% charge of one is transferred to another, then force between them will become

(1) Zero

(2) $2F$

(3) $\frac{3F}{4}$

(4) $\frac{F}{4}$

17. **Statement I:** Total charge of an isolated system remains conserved.
Statement II: Charge is a scalar quantity.

(1) Both statements I and II are true and statement II is correct explanation of statement I

(2) Both statements I and II are true but statement II is not correct explanation of I

(3) Statement I is true and statement II is false

(4) Both Statements I and II are false

18. A hydraulic lift, used at a service station, has areas of large and small pistons in ratio 5 : 1. Weight required on small piston which will be sufficient to lift a car of mass 1000 kg is

(1) 5 kg wt

(2) 10 kg wt

(3) 20 kg wt

(4) 200 kg wt

19. In a stream line flow,
a. The speed of particle always remains same
b. The velocity of the particle always remains same.
c. The kinetic energies of all the particles of same mass arriving at a given point are same.
d. The momentum of all the particles of same mass arriving at the given point are same.
Choose the correct statement(s) from the options given below.

(1) a and b only

(2) b and d only

(3) c and d only

(4) All a, b, c and d

20. A single spherical drop falling through air with terminal speed 20 cm/s splits into eight small identical spherical droplets. The terminal speed of small droplet is

(1) 20 cm/s

(2) 10 cm/s

(3) 5 cm/s

(4) 1.5 cm/s

21. Two spherical rain drops with radii in the ratio 1:2 fall with a constant velocity in atmosphere. The ratio of their linear momenta in this situation is

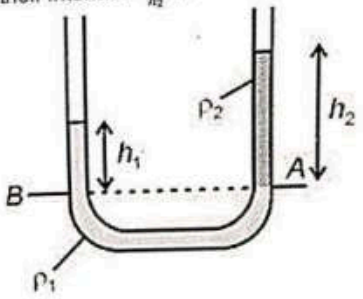
(1) 1 : 8

(2) 1 : 2

(3) 1 : 32

(4) 2 : 1

22. Two immiscible liquids are placed in a *U*-tube having densities ρ_1 and ρ_2. The ratio of height of the liquids above their interface $\frac{h_1}{h_2}$ is



(1) $\frac{\rho_1}{\rho_2}$

(2) $\frac{\rho_2}{\rho_1}$

(3) $\rho_1 \rho_2$

(4) $\frac{1}{\rho_1 \rho_2}$

23. On heating a cylindrical metallic rod, its length increases by 2%. Then area of cross-section of rod will

(1) Decrease by 2%

(2) Increase by 2%

(3) Decrease by 4%

(4) Increase by 4%

24. 5 kg of water at 10°C is added to 5 kg water at 40°C. Neglecting heat capacity of vessel and other heat losses. The equilibrium temperature is

(1) 4°C

(2) 35°C

(3) 25°C

(4) 33°C

25. A body cools from 80°C to 60°C in 2 minute. Time in which it will cool from 60°C to 40°C is (Temperature of surroundings is 10°C)

(1) 3 minute

(2) 2 minute

(3) 4 minute

(4) 5 minute

26. If temperature of an object in Fahrenheit scale is 104°F, then its temperature in centigrade scale is

(1) 40°C

(2) 136°C

(3) 76°C

(4) 104°C

27. A : When the temperature of a black body is doubled from *T* K to 2*T* K, the radial power becomes 16 times.
R : The radiant power of a body is proportional to fourth power of absolute temperature.

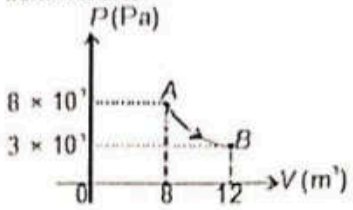
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28. 0.5 moles of an ideal monoatomic gas undergoes a thermodynamic process from A to B as shown in figure. The change in internal energy of the gas during the process from A to B is



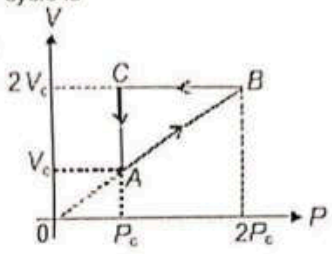
(1) – 7 kJ

(2) – 21 kJ

(3) – 28 kJ

(4) – 42 kJ

29. A hypothetical ideal gas with adiabatic exponent $\gamma = 3$, undergoes cyclic process as shown figure. The efficiency of cycle is



(1) $\frac{1}{6}$

(2) $\frac{1}{9}$

(3) $\frac{1}{2}$

(4) $\frac{1}{3}$

30. Internal energy of an ideal gas is given by the relation $U = \frac{3PV}{2}$. At constant volume, state of gas changes from (P_0, V_0) to $(3P_0, V_0)$, then

(1) Heat rejected by gas is P_0V_0

(2) Heat absorbed by gas is $\frac{3P_0V_0}{2}$

(3) Heat absorbed by gas is $3P_0V_0$

(4) Work done by gas is $2P_0V_0$

31. The ratio of change in internal energy to the heat interaction of an ideal monoatomic gas in the isobaric process will be

(1) $\frac{2}{5}$

(2) $\frac{3}{5}$

(3) $\frac{2}{3}$

(4) $\frac{3}{2}$

32. Consider the following statements
A. Work and heat are path functions in thermodynamics.
B. The internal energy of a gaseous system is state function.
C. For gaseous system, C_P is greater than C_V.
D. Work done by gas at constant volume is zero.
Based on above information pick the correct option.

(1) Only statement (A) is correct

(2) Only statements (A), (B) and (C) are correct

(3) Only statements (B), (C) and (D) are correct

(4) All statements (A), (B), (C) and (D) are correct

33. The speeds of ten particles (in m/s) are 2, 3, 3, 4, 5, 5, 5, 8, 9, 9. The most probable speed is

(1) 9 m/s

(2) 3 m/s

(3) 5 m/s

(4) 8 m/s

34. If two monoatomic gases have the same temperature, which of the following statements is true?

(1) Both gases have the same total kinetic energy

(2) Both gases have the same root mean square speed

(3) Both gases have equal density

(4) Lighter gas molecule move faster than heavier gas molecule

35. If M is the molar mass of a gas, then the average speed of its molecules at temperature T is

(1) $\sqrt{\frac{3RT}{\pi M}}$

(2) $\sqrt{\frac{8RT}{\pi M}}$

(3) $\sqrt{\frac{2RT}{M}}$

(4) $\sqrt{\frac{3RT}{M}}$

36. The displacement x of a particle executing simple harmonic motion is given as $x = 4\sin(\pi + 2t)$ m. Maximum speed of the particle is

(1) 4 m/s

(2) 2 m/s

(3) 8 m/s

(4) 8π m/s

37. A particle is executing SHM with time period T. If it starts its motion from mean position, then its velocity at $\frac{T}{12}$ second will be

(Where $T = \frac{2\pi}{\omega}$ and A is amplitude of the particle)

(1) $\frac{A\omega}{4}$

(2) $\frac{\sqrt{3}A\omega}{2}$

(3) $\frac{A\omega}{2}$

(4) $\frac{\sqrt{3}A\omega}{4}$

38. Two identical point masses P and Q, suspended from two separate massless springs of spring constants k_1 and k_2, respectively, oscillate vertically. If their maximum speeds are the same, the ratio (A_Q/A_P) of the amplitude A_Q of mass Q to the amplitude A_P of mass P is

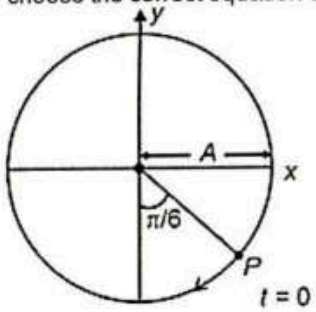
(1) $\frac{k_1}{k_2}$

(2) $\sqrt{\frac{k_2}{k_1}}$

(3) $\sqrt{\frac{k_1}{k_2}}$

(4) $\frac{k_2}{k_1}$

39. For the given phasor diagram for particle P under SHM, choose the correct equation of SHM



(1) $x = A\sin\left(\omega t + \frac{5\pi}{6}\right)$

(2) $x = A\sin\left(\omega t + \frac{\pi}{3}\right)$

(3) $x = A\cos\omega t$

(4) $x = 2A\sin(\omega t + \pi)$

40. For a spring-block system performing SHM, at a certain position, kinetic energy of block and elastic potential energy of spring are both equal to 16 J. If stiffness of spring is 1 N/m, then amplitude of SHM is (Assume zero potential energy at mean position)

(1) 4 m

(2) $4\sqrt{2}$ m

(3) 8 m

(4) 16 m

41. The two nearest harmonics of an open organ pipe are 300 Hz and 450 Hz. If speed of sound in the pipe is 300 m/s, then length of the pipe is

(1) 500 cm

(2) 75 cm

(3) 150 cm

(4) 100 cm

42. If at $t = 0$, a travelling wave pulse on a string is described by the function $y = \frac{6}{x^2+3}$. What will be the wave function representing the pulse at time t, if the pulse is propagating along positive x-axis with speed 4 m/s?

(1) $y = \frac{6}{(x+4t)^2+3}$

(2) $y = \frac{6}{(x-4t)^2+3}$

(3) $y = \frac{6}{(x+t)^2}$

(4) $y = \frac{6}{(x-t)^2+10}$

43. A wave travelling along the x-axis is described by the equation $y(x, t) = 0.1\sin(Px - Qt)$. If the wavelength and time period of wave are 0.06 m and 1 s, respectively, then value of P and Q in appropriate unit will be

(1) $P = 2\pi$, $Q = \frac{100\pi}{3}$

(2) $P = \frac{100\pi}{3}$, $Q = 2\pi$

(3) $P = \pi$, $Q = \frac{50\pi}{3}$

(4) $P = \frac{50\pi}{3}$, $Q = \pi$

44. The displacement y in x direction is given by $y = 10^{-2}\sin\left(300t - 4x + \frac{\pi}{6}\right)$ m, where x is in meter, t in s, then the speed of the wave motion is

(1) 300 m/s

(2) 200 m/s

(3) 75 m/s

(4) 150 m/s

45. The equation of a simple harmonic plane progressive wave is given by $y = 5\sin\pi(2t - 0.02x)$ where t in second and x in cm. What will be the phase difference of two medium particles separated by a distance 100 cm at any time t?

(1) π

(2) $\frac{\pi}{6}$

(3) 2π

(4) $\frac{2\pi}{3}$

CHEMISTRY

46. The pair which represents intensive properties is

(1) Volume and temperature

(2) Enthalpy and entropy

(3) Heat capacity and internal energy

(4) Density and pressure

47. Consider the following statements
(a) On boiling of egg entropy increases.
(b) On stretching of rubber entropy decreases.
(c) During evaporation of water entropy decreases.
Correct statement(s) is/are

(1) (b) only

(2) (a) and (b) only

(3) (a) and (c) only

(4) (a), (b) and (c)

48. An ideal gas is allowed to expand isothermally against a constant external pressure of 2.5 atm from an initial volume of 2.5 L to final volume of 4.5 L. The change in internal energy for this process is

(1) –505 J

(2) 505 J

(3) 113.625 J

(4) 0 J

49. If a process is spontaneous at only low temperature, then

(1) $\Delta H < 0$ and $\Delta S > 0$

(2) $\Delta H > 0$ and $\Delta S > 0$

(3) $\Delta H < 0$ and $\Delta S < 0$

(4) $\Delta H > 0$ and $\Delta S < 0$

50. 2 mole of an ideal gas at 127°C undergoes expansion isothermally and reversibly from 1 litre to 10 litre. The entropy change in the process is

(1) 38.29 J/K

(2) 32.94 J/K

(3) 46.34 J/K

(4) 44.44 J/K

51. The enthalpy of formation of XY is –100 kJ mol^{-1}. If the bond dissociation enthalpy of X_2 and Y_2 are 20 and 40 kJ mol^{-1} respectively then bond dissociation enthalpy of XY (in kJ mol^{-1}) is

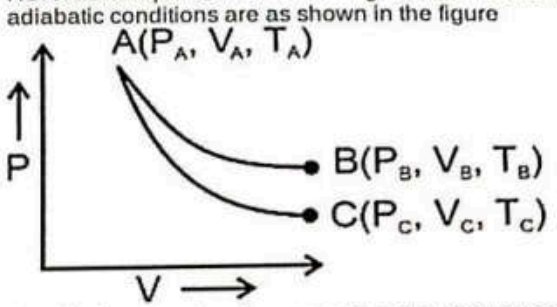
(1) 70

(2) 200

(3) 50

(4) 130

52. Reversible expansion of an ideal gas under isothermal and adiabatic conditions are as shown in the figure



Identify the correct option regarding the above-mentioned P – V curve.

(1) AB is an adiabatic expansion process

(2) $\Delta S_{AB} > \Delta S_{AC}$

(3) AC an isothermal expansion process

(4) $T_C = T_B$

53. Match list-I with list-II and choose the correct option.

	List-I (Reactions)	**List-II (Relation between ΔH and ΔU)**
a.	$N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$	(i) $\Delta H = \Delta U - \frac{1}{2}RT$
b.	$PCl_5(g) \rightarrow PCl_3(g) + Cl_2(g)$	(ii) $\Delta H = \Delta U + RT$
c.	$C(s) + O_2(g) \rightarrow CO_2(g)$	(iii) $\Delta H = \Delta U - 2RT$
d.	$CO(g) + \frac{1}{2}O_2(g) \rightarrow CO_2(g)$	(iv) $\Delta H = \Delta U$

(1) a(iii), b(iv), c(i), d(ii)

(2) a(ii), b(iii), c(iv), d(i)

(3) a(ii), b(iv), c(i), d(iii)

(4) a(iii), b(ii), c(iv), d(i)

54. 100 ml 0.2 M NaOH solution is mixed with 75 ml 0.4 M HCl solution. The heat evolved in the neutralization process is

(1) 325 cal

(2) 125 cal

(3) 274 cal

(4) 13.7 kcal

55. 1 mole of an ideal gas at 300 K and 10 atm is expanded to 1.0 atm pressure isothermally and reversibly then work involved is

(1) 5.74 kJ

(2) 5744 kJ

(3) 57.44 kJ

(4) 5.7 J

56. Conjugate acid and conjugate base of HCO_3^- respectively are

(1) $H_2C_2O_4$ and CO_2

(2) CO_3^{2-} and H_2CO_3

(3) CO_2 and CO_3^{2-}

(4) H_2CO_3 and CO_3^{2-}

57. For the reaction A(g) $\rightleftharpoons$ 2B(g), the backward reaction rate constant is higher than the forward reaction rate constant by a factor of 2500, at 1000 K.
[Given : R = 0.0831 L atm mol^{-1} K^{-1}]
K_P for the reaction at 1000 K is

(1) 0.033

(2) 0.021

(3) 83.1

(4) 2.077×10^5

58. Higher yield of NO in $N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$ can be obtained at
[ΔH of the reaction = +180.7 kJ mol^{-1}]
A. Higher temperature
B. Lower temperature
C. Higher concentration of N_2
D. Higher concentration of O_2
Choose the **correct** answer from the options given below :

(1) B, C, D only

(2) A, C, D only

(3) A, D only

(4) B, C only

59. The K_{sp} of Ag_2CrO_4 is 1.1×10^{-12} at 298 K. The solubility of Ag_2CrO_4 in a 0.2 M $AgNO_3$ solution is (in mol/L)

(1) 2.75×10^{-11}

(2) 5.5×10^{-12}

(3) 1.5×10^{-10}

(4) 6.5×10^{-10}

60. The percent ionisation of 0.02 M formic acid in 0.1 M HCl will be nearly (K_a of formic acid is 1.8×10^{-4})

(1) 0.07%

(2) 0.09%

(3) 0.35%

(4) 0.18%

61. Consider the following statements.
Statement I: The solution which resist change in pH on dilution or with addition of small amount of acid or base is called buffer solution.
Statement II: A mixture of benzoic acid and sodium benzoate acts as buffer solution.
In the light of above statements, choose the **correct** option.

(1) Statement I is incorrect but statement II is correct

(2) Statement I is correct but statement II is incorrect

(3) Both statement I and II are correct

(4) Both statement I and II are incorrect

62. Which among the following does not undergo hydrolysis?

(1) KCl

(2) NH_4Cl

(3) CH_3COONa

(4) CH_3COONH_4

63. In which of the following cases, the pH is less than 7 at 25°C.

(1) 50 mL of 0.1 M HNO_3 + 50 mL of 0.1 M $Ca(OH)_2$

(2) 50 mL of 0.1 M CH_3COOH + 50 mL of 0.1 M KOH

(3) 50 mL of 0.1 M H_2SO_4 + 50 mL of 0.2 M NaOH

(4) 50 mL of 0.1 M H_2SO_4 + 50 mL of 0.1 M NaOH

64. The molar solubility of AgCl in 0.1 M solution of NaCl is (K_{sp} of AgCl = 1.8×10^{-10})

(1) 1.8×10^{-8} M

(2) 1.8×10^{-9} M

(3) 1.34×10^{-5} M

(4) 1.8×10^{-5} M

65. Given below are two statements
Statement I: Oxidation state of terminal Br in Br_3O_8 is +7.
Statement II: Average oxidation number of Br in Br_3O_8 is $\frac{+16}{3}$.
In the light of above statements, choose the most appropriate answer from the options given below.

(1) Statement I is correct but statement II is incorrect

(2) Statement I is incorrect but statement II is correct

(3) Both statement I and statement II are correct

(4) Both statement I and statement II are incorrect

66. Consider the following reaction
(i) $P_4 + 3OH^- \rightarrow PH_3 + 3H_2PO_2^-$
(ii) $2H_2O_2 \rightarrow 2H_2O + O_2$
(iii) $Cl_2 + 2OH^- \rightarrow ClO^- + Cl^- + H_2O$
The disproportionation reaction is/are

(1) (i) and (ii) only

(2) (iii) only

(3) (ii) and (iii) only

(4) (i), (ii) and (iii)

67. Given below are the two statements
Statement I: Species that loses electron is called reducing agent during a redox reaction.
Statement II: Species that gains electron is called oxidising agent during a redox reaction.
In the light of above statements, choose the most appropriate answer from the options given below

(1) Statement I is correct but statement II is incorrect

(2) Statement I is incorrect but statement II is correct

(3) Both statement I and statement II are correct

(4) Both statement I and statement II are incorrect

68. Which among the following will not liberate H_2 gas with dilute H_2SO_4?

(1) Cu

(2) Na

(3) Fe

(4) Al

69. Which among the following is best reducing agent in aqueous medium?

(1) Na

(2) Li

(3) Cu

(4) F_2

70. Consider the following redox reaction
$aCr_2O_7^{2-}(aq) + bSO_3^{2-}(aq) + cH^+(aq) \rightarrow dCr^{3+}(aq) + eSO_4^{2-}(aq) + fH_2O(l)$
If a, b, c, d, e and f represents the coefficients used for balancing given reaction then **correct** ratio of a : b : c is

(1) 2 : 3 : 7

(2) 1 : 8 : 3

(3) 1 : 3 : 8

(4) 2 : 3 : 8

71. Correct order of freezing point of the given aqueous solution is
(i) 0.2 m NaCl (α = 50%)
(ii) 0.2 m $Al_2(SO_4)_3$ (α = 20%)
(iii) 0.2 m Na_3PO_4 (α = 40%)

(1) (iii) > (i) > (ii)

(2) (i) > (ii) > (iii)

(3) (iii) > (ii) > (i)

(4) (i) > (iii) > (ii)

72. Match List-I with List-II

List-I (Example)	List-II (Type of Solution)
A. Humidity	I. Solid in solid
B. Alloys	II. Liquid in gas
C. Amalgams	III. Solid in gas
D. Smoke	IV. Liquid in solid

Choose the **correct** answer from the options given below:

(1) A-III, B-I, C-IV, D-II

(2) A-III, B-II, C-I, D-IV

(3) A-II, B-IV, C-I,D-III

(4) A-II, B-I, C-IV, D-III

73. Which of the following pairs will form an ideal solution?

(1) Chloroform and acetone

(2) Ethanol and acetone

(3) n-Hexane and n-Heptane

(4) Phenol and Aniline

74. 0.3 molal aqueous solution of a salt AX_2 is 40% ionized. If K_b for water is 0.52 K kg mol^{-1} then the boiling point of the solution will be

(1) 100.28°C

(2) 101.42°C

(3) 100.61°C

(4) 101.81°C

75. A 0.2 molal aqueous solution of electrolyte AB_2 freezes at –0.88°C. If K_f for H_2O is 1.86 K kg mol^{-1} then degree of dissociation of AB_2 is

(1) 0.52

(2) 0.68

(3) 0.76

(4) 0.84

76. Which among of the following gases is least soluble in water at 298 K?

Gases	K_H/kbar
Argon	40. 3
Carbon dioxide	1. 67
Formaldehyde	1.83×10^{-5}
Methane	0. 413

The **correct** answer is

(1) Argon

(2) Carbon dioxide

(3) Formaldehyde

(4) Methane

77. Choose the incorrect statement

(1) Chloroform molecule is able to form hydrogen bond with acetone molecule

(2) Phenol and aniline mixture shows negative deviation from Raoult's law

(3) 60% aqueous ethanol solution forms minimum boiling azeotrope at one atm pressure

(4) Mixture of ethanol and acetone shows positive deviation from Raoult's law

78. Given below are two statements one is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A) : Aquatic species are more comfortable in cold water rather than in warm water.
Reason (R) : Solubility of gases increases with increase in temperature.
In the light of the above statements, choose the correct answer.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) Both (A) and (R) are false

79. One molal solution of a carboxylic acid in benzene shows the elevation of boiling point of 1.518 K. The degree of association for dimerisation of the acid in benzene is (K_b for benzene is 2.53 K kg mol^{-1})

(1) 60%

(2) 70%

(3) 75%

(4) 80%

80. Correct order of osmotic pressure of given solution is
A. 0.1 M sucrose
B. 0.2 M glucose
C. 0.01 M NaCl
D. 0.05 M $CaCl_2$

(1) B > D > A > C

(2) B > A > C > D

(3) D > C > A = B

(4) B > C = D > A

81. Consider the following two statements.
Statement (I): Galvanic cells that are designed to convert the energy of combustion of fuels like hydrogen, methane, etc directly into electrical energy are called fuel cells.
Statement (II): A secondary cell after use can be recharged by passing current through it in the opposite direction.
In the light of above two statements, choose the correct option.

(1) Both statement (I) and statement (II) are correct

(2) Statement (I) is correct but statement (II) is incorrect

(3) Statement (I) is incorrect but statement (II) is correct

(4) Both statement (I) and statement (II) are incorrect

82. The conductivity of 0.1 M solution of KCl at 298 K is 0.0248 S cm^{-1}, the molar conductivity of the given solution will be

(1) 213 S cm^2 mol^{-1}

(2) 62 S cm^2 mol^{-1}

(3) 248 S cm^2 mol^{-1}

(4) 124 S cm^2 mol^{-1}

83. The standard Gibbs energy of the given cell reaction will be
$3Ni(s) + 2Au^{3+}(aq) \rightarrow 3Ni^{2+}(aq) + 2Au(s)$
(Given: $E^\circ_{Ni^{2+}/Ni} = -0.25$ V and $E^\circ_{Au^{3+}/Au} = 1.4$ V)

(1) – 477.67 kJ mol^{-1}

(2) – 955.35 kJ mol^{-1}

(3) – 318.45 kJ mol^{-1}

(4) – 636.9 kJ mol^{-1}

84. If the molar conductivity (Λ_m) of a 0.050 mol L^{-1} solution of a monobasic weak acid is 90 S cm^2 mol^{-1}, its extent (degree) of dissociation will be
[Assume Λ°_+ = 349.6 S cm^2 mol^{-1} and Λ° = 50.4 S cm^2 mol^{-1}.]

(1) 0.225

(2) 0.215

(3) 0.115

(4) 0.125

85. If standard reduction potential of A^{2+}/A, B^{2+}/B and C^{2+}/C are 0.52 V, –0.14 V and – 2.93 V respectively, then reducing power of the metals will be in order of

(1) A > C > B

(2) A > B > C

(3) B > A > C

(4) C > B > A

86. In the electrolysis of aqueous NaCl solution the products obtained at anode and cathode respectively are.

(1) O_2 and H_2

(2) Cl_2 and H_2

(3) H_2 and O_2

(4) HCl and O_2

87. A solution of $CuSO_4$ is electrolysed for 90 minutes with a current of 4.825 amperes. The mass of copper deposited at the cathode is
(Atomic mass of copper = 63.5 u)

(1) 12.4 g

(2) 17.2 g

(3) 8.6 g

(4) 6.2 g

88. Standard cell potential for a cell represented as $Cu(s)|Cu^{2+}(0.01M)||Cu^{2+}(0.001M)|Cu(s)$ will be

(1) 0.029 V

(2) – 0.029 V

(3) 0.059 V

(4) 0

89. Match List-I with List-II

List-I (Type of cells)	List-II (Reaction at anode)
a. Leclanche cell	(i) $2H_2(g) + 4OH^-(aq) \rightarrow 4H_2O(l) + 4e^-$
b. Lead storage cell (battery is in use)	(ii) $Zn(Hg) + 2OH^- \rightarrow ZnO(s) + H_2O + 2e^-$
c. Mercury cell	(iii) $Pb(s) + SO_4^{2-}(aq) \rightarrow PbSO_4(s) + 2e^-$
d. $H_2 - O_2$ fuel cell	(iv) $Zn(s) \rightarrow Zn^{2+} + 2e^-$

Choose the correct answer from the options given below

(1) a(iii), b(ii), c(iv), d(i)

(2) a(iv), b(iii), c(ii), d(i)

(3) a(ii), b(iii), c(i), d(iv)

(4) a(ii), b(iii), c(iv), d(i)

90. Given below are two statements
Statement I: In corrosion, at anode, Fe gets oxidised to Fe^{2+} ion.
Statement II: In corrosion, at cathode, $O_2(g)$ reduced in presence of H^+ ions into H_2O.
In the light of above statements, choose the correct answer from the options given below.

(1) Both statement I and statement II are true

(2) Statement I is true but statement II is false

(3) Statement I is false but statement II is true

(4) Both statement I and statement II are false

BOTANY

91. Select the **correct** match w.r.t 'aestivation' of petals.

(1) Valvate – Pea

(2) Twisted – *Cassia*

(3) Imbricate – Cotton

(4) Vexillary – Bean flower

92. Stem tendrils are **not** found in

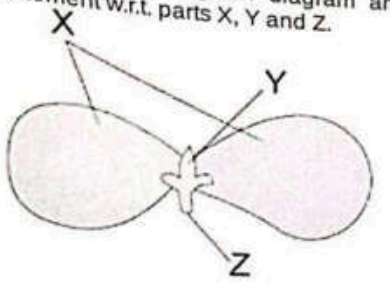
(1) Pumpkin

(2) Pea

(3) Watermelon

(4) Cucumber

93. Refer the below given diagram and select the **correct** statement w.r.t. parts X, Y and Z.



(1) 'Z' represents the structure that develops into shoot system on germination of seed

(2) 'Y' represents the structure that is enclosed in coleorhiza

(3) In grasses, the structure 'X' is called as coleoptile

(4) 'X' represents the structures that are often fleshy and full of reserve materials

94. Fruits of which of the given plants have persistent calyx?

(1) Brinjal

(2) Mustard

(3) Tulip

(4) *Gloriosa*

95. The tap roots of carrot modify for the purpose of

(1) Mechanical support

(2) Storage of food

(3) Respiration

(4) Vegetative propagation

96. The swollen leaf base in legumes is called

(1) Stipule

(2) Lamina

(3) Leaf blade

(4) Pulvinus

97. Diadelphous condition in pea shows that

(1) Out of total stamens, only two stamens are fused together

(2) Total two stamens are present

(3) Stamens are united into two bundles

(4) Both types of sex organs are present

98. How many of the features are associated with mustard plant?

(a) Actinomorphic flower
(b) Pentamerous flower
(c) Superior ovary
(d) Vexillary aestivation of corolla
(e) Epipetalous stamens
(f) Parietal placentation

(1) Three

(2) Four

(3) Five

(4) Six

99. Match the following columns.

Column I		Column II
a. Epiphyllous	(i)	Stamens attached to the petals
b. Apocarpous	(ii)	Stamens attached to the perianth
c. Monoadelphous	(iii)	Carpels are free
d. Epipetalous	(iv)	Stamens are united in a single bundle

(1) a(iv), b(iii), c(ii), d(i)

(2) a(ii), b(iii), c(iv), d(i)

(3) a(iv), b(i), c(iii), d(ii)

(4) a(ii), b(iv), c(iii), d(i)

100. Fibrous root system differs from adventitious root system as

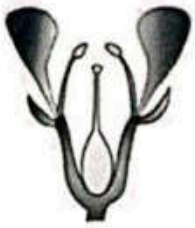
(1) It replaces primary root which is short lived

(2) It can develop from any part of the plant other than radicle

(3) It can be above the ground also

(4) It is seen in dicots only

101. Which of the following represents the position of floral parts on thalamus in Mustard and China rose?

(1) 

(2) 

(3) 

(4) 

102. The cells proximal to the meristematic region of root are

(1) Thin-walled, small, with dense protoplasm and suberin

(2) Responsible for the growth of roots in length

(3) Gradually differentiated and have root cap

(4) Responsible for continuous division and have root hairs

103. Two plants A and B belonging to different families, show following features

Plant A - Zygomorphic, bisexual flower with non-endospermic seed.

Plant B - Persistent calyx, epipetalous stamen and axile placentation.

Identify the plants.

(1) A – *Aloe*, B – Mustard

(2) A – Tomato, B – *Indigofera*

(3) A – *Sesbania*, B – *Asparagus*

(4) A – Soyabean, B – Tobacco

104. Flower has radial symmetry in

(1) Chilli

(2) *Cassia*

(3) Bean

(4) Canna

105. The drupe fruits are developed from

(1) Bicarpellary gynoecium

(2) Monocarpellary gynoecium

(3) Apocarpous ovary

(4) Syncarpous ovary

106. $\oplus\ \female\ K_{(5)}\ \overset{\frown}{C}_{(5)}\ A_5\ \underline{G}_{(2)}$

is the floral formula of a plant family whose one of the members is

(1) *Petunia*

(2) *Gloriosa*

(3) *Indigofera*

(4) *Trifolium*

107. In grasses, guard cells of stomata are

(1) Bean-shaped

(2) Kidney-shaped

(3) Sickle shaped

(4) Dumb-bell shaped

108. The feature which is **not** associated with internal structure of a dicotyledonous leaf is

(1) Closed and conjoint vascular bundle

(2) Palisade parenchyma found on the adaxial side

(3) Same sized vascular bundles that does not depend upon vein size

(4) Presence of more stomata on the abaxial epidermis generally

109. Bulliform cells are

(1) Present mainly in dicot stems

(2) Photosynthetic in nature

(3) Colorless and empty

(4) Dead and highly lignified

110. The structure which is usually absent in monocots, is

(1) Phloem parenchyma

(2) Xylem vessels

(3) Sieve tube elements

(4) Tracheids

111. Stele is constituted by all, **except**

(1) Pith

(2) Vascular bundles

(3) Endodermis

(4) Pericycle

112. Match the column I with column II and select the **correct** option.

Column I	Column II
a. Pericycle absent	(i) Dicot stem
b. Ring like arrangement of vascular bundles	(ii) Monocot root
c. Small and inconspicuous pith	(iii) Monocot stem
d. Xylem bundles are said to be polyarch	(iv) Dicot root

(1) a(iv), b(iii), c(ii), d(i)

(2) a(iii), b(i), c(ii), d(iv)

(3) a(ii), b(iii), c(iv), d(i)

(4) a(iii), b(i), c(iv), d(ii)

113. Conjoint open vascular bundle is found in

(1) Dicot root

(2) Dicot stem

(3) Monocot root

(4) Monocot stem

114. What is **wrong** about trichomes?

(1) They may be secretory in some plants

(2) Mainly help in preventing the water loss due to transpiration

(3) They are usually multicellular

(4) They help in gaseous exchange

115. The parenchymatous cells which lie between the xylem and phloem, seen in a section of monocot root are called

(1) Medullary rays

(2) Bast fibres

(3) Conjunctive tissues

(4) Secondary tissues

116. The outermost layer in transverse section of dicot root is

(1) Cortex

(2) Endodermis

(3) Pericycle

(4) Epiblema

117. Mark the following statements as true **(T)** or false **(F)** and select the **correct** option.

A. Phloem fibres of the flax are of commercial importance.

B. Root epidermis is covered with cuticle and contains root hairs.

C. Epidermis is always single layered.

	A	B	C
(1)	T	T	F
(2)	F	F	F
(3)	T	F	F
(4)	F	T	F

(1) (1)

(2) (2)

(3) (3)

(4) (4)

118. Identify the various layers of tissue which constitute the ground tissue system, by considering the following features of different zones in a dicot stem. Arrange them in a **correct** order, *i.e.*, from periphery to the centre and choose the option accordingly

A. A single layer of compactly arranged cells which contain starch grains.

B. Semilunar patches of sclerenchyma just below the endodermis.

C. Few layers of collenchymatous cells present just below the epidermis providing mechanical strength to the young stem.

D. Innermost part of the stele

(1) B → C → D → A

(2) A → C → B → D

(3) C → A → B → D

(4) C → B → A → D

119. Find the **mismatched** pair.

(1) Vessels : Not found in most of the gymnosperms

(2) Tracheids : Dead and without protoplasm

(3) Xylem fibre : Living and thin walled

(4) Xylem parenchyma : Stores tannins

120. Casparian strips are present in the endodermis of

(1) Dicot stem

(2) Dicot root

(3) Monocot stem

(4) Dicot leaf

121. Read the following statements and choose the **correct** option.
(A) The members of chlorophyceae are usually grass green in colour due to dominance of pigments chlorophyll 'a' and 'b'.
(B) In red algae, complex post-fertilisation developments occur

(1) Only statement (A) is correct

(2) Only statement (B) is correct

(3) Both statements (A) and (B) are correct

(4) Both statements (A) and (B) are incorrect

122. Select the **correct** statement w.r.t. first embryophytic amphibians of plant kingdom.

(1) Sperm mother cell produces biflagellated antherozoids

(2) They are heterosporous

(3) Vessels are absent in xylem

(4) They are vascular terrestrial plants

123. *Porphyra* differs from *Fucus* in

(1) Absence of sexual reproduction

(2) Absence of cellulose in cell wall

(3) Photosynthetic pigment of reaction centre in photosystem

(4) Nature of male gametes and spores

124. Prothallus of a fern is

(1) Gametophytic

(2) Non photosynthetic

(3) Conspicuous

(4) Unicellular

125. The leaves of gymnosperms are well adapted to withstand extremes of environmental conditions due to

(1) Being broad and hard

(2) Presence of thick cuticle and sunken stomata

(3) Presence of superficial stomata

(4) Presence of transfusion tissue

126. The spread of living pteridophytes is limited and restricted to narrow geographical regions because of

(1) Dependence of gametophytes on sporophytes

(2) Gametophytes that require cool, damp and shady places to grow

(3) No need of water for fertilisation

(4) Sporophyte which can grow only in dry conditions

127. Identify true **(T)** or false **(F)** for the following statements and then choose the **correct** option.
(a) Mosses along with lichens are the first organisms to colonise rocks.
(b) Protonema is a creeping, green, branched and frequently filamentous stage of liverworts.
(c) In gymnosperms male and female gametophytes are present within sporangia located on sporophyte.
(d) In brown algae gametes are pear shaped and bear two laterally attached flagella.

	(a)	(b)	(c)	(d)
(1)	T	T	T	T
(2)	T	F	T	F
(3)	T	F	T	T
(4)	F	T	F	T

(1) (1)

(2) (2)

(3) (3)

(4) (4)

128. Artificial system of classification given by Linnaeus

(1) Was a phylogenetic classification system

(2) Was based upon both external and internal features

(3) Gave equal weightage to vegetative and sexual characteristics

(4) Was based upon the natural affinities among the organisms

129. Agar and algin are obtained from

(1) *Gracilaria* and *Gelidium* respectively

(2) Brown algae and red algae respectively

(3) The members of Rhodophyceae and Phaeophyceae respectively

(4) Green algae and brown algae respectively

130. Vascular tissues are absent in algae. One of the reasons is that

(1) Most of them are unicellular

(2) All the cells of the plant body are photosynthetic

(3) Main plant body is generally haploid

(4) They are aquatic and so water conduction is not required

131. Fusion between a large non-motile female gamete and a smaller motile male gamete is found in

(1) *Ulothrix*

(2) *Spirogyra*

(3) *Polysiphonia*

(4) *Volvox*

132. Which of the following options depicts the **correct** sequence of development of different gametophytic stages of mosses?

(1) Primary Protonema → Spore → Leafy stage → Secondary protonema

(2) Spore → Primary Protonema → Secondary protonema → Leafy stage

(3) Spore → Leafy stage → Primary Protonema → Secondary protonema

(4) Leafy stage → Spore → Secondary protonema → Primary Protonema

133. Match List I with List II

	List-I		List-II
a.	*Pinus*	(i)	Elaborate mechanism of spore dispersal, present in capsule
b.	*Polysiphonia*	(ii)	The female gametophytes are retained on the parent sporophytes for variable periods
c.	*Funaria*	(iii)	Male and female gametophytes do not have an independent free-living existence
d.	*Salvinia*	(iv)	Complex post fertilisation developments and lacks embryo

Choose the **correct** option

(1) a(ii), b(i), c(iii), d(iv)

(2) a(iii), b(ii), c(iv), d(i)

(3) a(iii), b(iv), c(i), d(ii)

(4) a(ii), b(i), c(iv), d(iii)

134. Read the following statements and select the **correct** option.

Assertion (A) : Being photosynthetic, algae increase the level of dissolved oxygen in their immediate environment.

Reason (R) : Algae are the primary producers of energy rich compounds which form the basis of food cycles of all aquatic animals.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

135. Strobili or cones are found in

(1) *Selaginella*, *Equisetum*

(2) *Selaginella*, *Funaria*

(3) *Equisetum*, *Funaria*

(4) *Selaginella*, *Sphagnum*

ZOOLOGY

136. Which of the following statements is **incorrect** w.r.t. counter current mechanism?

(1) NaCl is transported by ascending limb of Henle's loop which is exchanged with the descending limb of vasa recta.

(2) NaCl is returned to the medullary interstitium by the ascending portion of vasa recta.

(3) Small amounts of urea enter thin segment of the ascending limb of Henle's loop which is transported back to the interstitium by the collecting duct.

(4) Ascending limb of vasa recta is found in close proximity to ascending limb of loop of Henle and also have counter flow with it.

137. The descending limb of loop of Henle

(1) Dilutes the glomerular filtrate

(2) Concentrates the glomerular filtrate

(3) Is the site of minimum reabsorption

(4) Is highly permeable to electrolytes

138. Glomerular filtration occurs through filtration membrane which contains all of the following, **except**

(1) Epithelial cells of Bowman's capsule

(2) Endothelium of glomerular blood capillaries

(3) Epithelial cells of efferent arteriole

(4) Basement membrane

139. Match column I with column II w.r.t. a healthy adult human.

	Column I		Column II
a.	GFR	(i)	12 L/hr
b.	Amount of blood filtered by kidneys	(ii)	5 L/min
c.	Cardiac output	(iii)	180 L/day
d.	Amount of CO_2 removed from lungs	(iv)	1100-1200 mL/min

Select the correct option.

(1) a(iii), b(iv), c(ii), d(i)

(2) a(iv), b(ii), c(i), d(iii)

(3) a(iii), b(iv), c(i), d(ii)

(4) a(iv), b(iii), c(ii), d(i)

140. Read the following statements and choose the **correct** option

Statement A: Peritubular capillaries are absent in cortical nephrons.

Statement B: Cortical nephrons are not involved in urine formation.

(1) Both A and B statements are correct

(2) Both A and B statements are incorrect

(3) Only statement A is correct

(4) Only statement B is correct

141. If a person is suffering from dehydration, which of the following will cause restoration of blood volume and GFR?

(1) Release of renin from JG cells

(2) Release of a powerful vasodilator Angiotensin – II

(3) Release of aldosterone from adrenal medulla

(4) Release of ADH from JG apparatus

142. Choose the **mismatch**

(1)	Glycosuria	–	Presence of glucose in urine
(2)	Ketonuria	–	Presence of ketone bodies in urine
(3)	Uremia	–	Presence of urea in urine
(4)	Glomerulonephritis	–	Inflammation of glomeruli of kidney

(1) (1)

(2) (2)

(3) (3)

(4) (4)

143. Consider the given statements w.r.t. humans.

(a) The primary function of sweat is to remove waste from the body.

(b) Sweat glands never eliminate lactic acid.

(c) Sweat glands remove large amounts of urea and uric acid.

(d) The secretions of sebaceous glands provide a protective oily covering for the skin.

How many of the above given statements is/are **incorrect**?

(1) One

(2) Two

(3) Three

(4) Four

144. If hepatic failure occurs in a person, then, which of the following substances will increase in blood?

(1) Ammonia

(2) Urea

(3) Uric acid

(4) Water

145. Read the following statements and choose the correct answer.

A : Each kidney of an adult human measures 5-7 cm in length and 10-12 cm in width.

B : The Malpighian corpuscles, PCT and DCT of the nephron are situated in the cortical region of the kidney whereas the loop of Henle dips into the medulla.

(1) Both statements are correct

(2) Both statements are incorrect

(3) Only statement A is correct

(4) Only statement B is correct

146. Mechanism of muscle contraction is best explained by the sliding filament theory. Select the incorrect statement w.r.t. it.

(1) Breaking of cross bridge occurs when the myosin head binds to a new ATP molecule.

(2) Myosin head bound with ADP and Pi is considered as energised state.

(3) Formation of cross bridge occurs when Ca^{2+} level in the sarcoplasm is low and myosin head is not attached with ATP

(4) During muscle contraction, actin and myosin filaments do not shorten but actin filaments slide over myosin filaments

147. Choose the correct statement regarding the disorders of muscular or skeletal system.

(1) Rapid spasms in muscle due to high Ca^{2+} in body fluid causes tetany

(2) Osteoporosis is characterised by decrease in bone mass and higher chances of fractures with advancing age.

(3) Myasthenia gravis is a genetic disorder in which there is inhibition of sliding of myosin filaments

(4) Muscular dystrophy is an auto-immune disorder characterised by shortening of muscles.

148. Compared to the visceral muscle fibres, the skeletal muscle fibres

(1) Assist in the transport of food through the digestive tract by peristalsis

(2) Are involved in locomotory actions and changes of body postures

(3) Are striated in appearance and involuntary in nature

(4) Have a single central nucleus

149. The total number of facial bones in an adult human is

(1) Equal to the total number of true ribs

(2) Less than the number of thoracic vertebrae

(3) Thrice the number of vertebrochondral ribs

(4) Equal to the number of carpals in both forelimbs

150. Choose the correct representation w.r.t. number of bones of an adult human.

(1) Sternum + Coxal bones = 3 bones

(2) Coxal bones + Femur = 7 bones

(3) Vertebrochondral ribs + Hyoid = 8 bones

(4) Cranial bones + Auditory ossicles = 9 bones

151. When the muscle is in maximally contracted state, then which of the given events occur?
(a) Reduction in the length of 'I' band
(b) Widening of the 'A' band
(c) Disappearance of 'H' zone
(d) Utilisation of the energy released from hydrolysis of ATP into ADP + P_i by myosin
Choose the **correct** option

(1) a, b, c and d

(2) a, b and c

(3) b, c and d

(4) a, c and d

152. Match the column I with column II and select the **correct** option.

	Column I		Column II
(i)	Ankle bones	(a)	Help in articulation of limbs with axial skeleton
(ii)	Neural canal	(b)	14 in number
(iii)	Girdle bones	(c)	Central hollow portion in each vertebra
(iv)	Acromian process	(d)	Present in scapula

(1) (i) – b, (ii) – c, (iii) – a, (iv) – d

(2) (i) – b, (ii) – c, (iii) – d, (iv) – a

(3) (i) – b, (ii) – a, (iii) – c, (iv) – d

(4) (i) – a, (ii) – b, (iii) – c, (iv) – d

153. All true ribs are A attached to the thoracic vertebrae and B connected to the sternum with the help of hyaline cartilage.
Choose the option that **correctly** fill the blanks A and B respectively.

(1) Ventrally, Medially

(2) Dorsally, Ventrally

(3) Laterally, Ventrally

(4) Laterally, Dorsally

154. The type of movement that assists in locomotion of *Euglena* also assists in

(1) Streaming of protoplasm in *Amoeba*

(2) Passage of ova through the oviduct

(3) Maintenance of water current in the canal system of *Sycon*

(4) Capturing of prey by *Hydra*

155. Select the incorrect statement w.r.t. an adult human.

(1) The skull region articulates with the superior region of the vertebral column with the help of two occipital condyles

(2) Sternum is the flat bone on the ventral midline of the thorax

(3) Appendicular skeleton comprises 126 bones and consists of bones of the limbs along with their girdles

(4) Scapula is situated in the ventral part of the thorax between the second and the seventh ribs

156. **Assertion (A):** Protein synthesising granules, called the Nissl's granules, are absent in axons of neurons.
Reason (R): Nissl's granules are composed of RER and ribosomes.
In the light of above statements, choose the **correct** option.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true, (R) is false

(4) Both (A) and (R) are false

157. Three major regions *i.e.* 'X', 'Y' and 'Z' make up the brain stem. 'Y' and 'Z' are the parts of hindbrain. Identify 'X' and choose the incorrect statement w.r.t. it.

(1) 'X' is located between the thalamus/hypothalamus of the forebrain and pons of the hindbrain

(2) A canal called cerebral aqueduct passes through the 'X'

(3) The dorsal portion of 'X' consists of corpora quadrigemina

(4) 'X' contains centres which control respiration, cardiovascular reflexes and body temperature

158. Following are the statements about polarised state as well as process of impulse conduction through a nerve fibre.
a. Neurons are non-excitable cells because their membranes are in a polarised state.
b. Polarised state of a nerve fibre is due to the difference in permeability of Na^+ and K^+.
c. The electrical potential difference across the resting plasma membrane is called the action potential.
d. Resting potential of axolemma is restored at the site of excitation by rapid efflux of K^+.
Choose the **correct** answer from the options given below.

(1) a and b are correct

(2) b and c are correct

(3) c and d are correct

(4) b and d are correct

159. The part of the human brain that consists of fibre tracts which interconnect different regions of the brain and contains a centre that can moderate the functions of the respiratory rhythm centre is

(1) Medulla oblongata

(2) Cerebellum

(3) Pons

(4) Hypothalamus

160. Nodes of Ranvier in PNS are gaps present between

(1) A dendrite and axon

(2) Two oligodendrocytes

(3) Two myelin sheaths

(4) A synaptic knob and dendrite

161. _____ has very convoluted surface in order to provide the additional space for many more neurons.
Select the **correct** option to fill in the blank.

(1) Pons

(2) Hypothalamus

(3) Hippocampus

(4) Cerebellum

162. Ionic gradients across the polarised axonal membrane is maintained by

(1) Active transport of $3Na^+$ outward for $2K^+$ into the axoplasm

(2) Passive transport of $3Na^+$ outward for $2K^+$ into the axoplasm

(3) Active transport of $2K^+$ outward for $3Na^+$ into the axoplasm

(4) Passive transport of $2K^+$ outward for $3Na^+$ into the axoplasm

163. Schwann cells are not present in
a. Autonomic neural system
b. Somatic neural system
c. Central nervous system
Choose the **correct** option

(1) (a) and (c)

(2) (b) and (c)

(3) (a), (b) and (c)

(4) (c) only

164. Choose the **incorrect** statement w.r.t the limbic system of the human brain.

(1) It influences emotional behaviour

(2) It includes of amygdala and hippocampus

(3) It is situated in the outer region of the cerebral hemisphere

(4) Along with hypothalamus, it regulates sexual behaviour

165. In humans, the part of the brain which receives and integrates visual, tactile and auditory inputs is

(1) Forebrain

(2) Hindbrain

(3) Hypothalamus

(4) Midbrain

166. Both insulin and thyroxine arrive at an organ at the same time, latter causes an effect on the organ but former does not. What can be the possible explanation for this?
Select the most appropriate option.

(1) Thyroxine is a water soluble hormone whereas insulin is not

(2) Thyroxine is a local hormone whereas insulin is a circulating hormone

(3) Thyroxine inhibits the action of insulin

(4) The organ cells have receptors for thyroxine but not for insulin

167. An endocrine gland 'X' is a lobular structure located between lungs behind the sternum on the ventral side of aorta. 'X' secretes hormone called 'Y'.
Select the correct option w.r.t 'Y'

(1) Proteinaceous in nature

(2) Major role is in differentiation of B-lymphocytes

(3) Inhibits production of antibodies

(4) Predominantly responsible for humoral immune response mediated by T-lymphocytes

168. How many of the mentioned hormones are not secreted by the organised endocrine glands of our body?

PRL, T_4, ANF, Epinephrine, CCK, Testosterone, Erythropoietin, Thymosin

Choose the **correct** option.

(1) Three

(2) Four

(3) Five

(4) Two

169. Match the columns and select the **correct** option.

	Column I		Column II
a.	Gastrin	(i)	Inhibits gastric secretion
b.	Gastric inhibitory peptide	(ii)	Stimulates the secretion of pancreatic enzymes
c.	Secretin	(iii)	Stimulates the secretion of water and bicarbonate from pancreas
d.	Cholecystokinin	(iv)	Stimulates the secretion of HCl

(1) a(ii), b(iii), c(i), d(iv)

(2) a(iv), b(i), c(iii), d(ii)

(3) a(ii), b(i), c(iii), d(iv)

(4) a(i), b(ii), c(iv), d(iii)

170. Androgens regulate the male sexual behaviour and are also involved in all of the following functions, **except**

(1) Maturation and normal functioning of the male accessory sex organs

(2) Spermatogenesis

(3) Erythropoiesis

(4) Catabolic effects on protein and carbohydrate metabolism

171. In a 24-year old girl, abnormal development of nodules in thyroid gland was observed. Later, her thyroid gland was completely removed surgically to prevent the spread of malignant cells. Which of the following could be the consequence(s) of removal of her thyroid gland ?
(a) Dysregulation of BMR
(b) Imbalanced erythropoiesis
(c) Disturbed water and electrolyte balance in the body
Select the correct option.

(1) Only (a)

(2) Only (b) and (c)

(3) Only (b)

(4) (a), (b) and (c)

172. Which of the following hormones bind to membrane bound receptors and acts through second messengers?

(1) Aldosterone

(2) Androgens

(3) Estrogens

(4) Follicle stimulating hormone

173. A teenager went for forest trekking and while walking, he suddenly came across a venomous snake. At this point, which of the following can be observed in his body?
a. Activation of sympathetic nervous system and release of catecholamines
b. Activation of parasympathetic nervous system and release of catecholamines
c. Pupillary dilation, breakdown of glycogen and hyperglycaemia in blood
Choose the **correct** option.

(1) a and c only

(2) b and c only

(3) a and b only

(4) a, b and c

174. Which of the following are **not** the effects of parathyroid hormone?

(a) Stimulates the process of bone resorption

(b) Decreases Ca^{2+} levels in blood

(c) Reabsorption of Ca^{2+} by renal tubules

(d) Decreases the absorption of Ca^{2+} from digested food

(e) Increases metabolism of carbohydrates

Choose the **most appropriate** answer from the options given below:

(1) (a) and (e) only

(2) (b) and (c) only

(3) (a) and (c) only

(4) (b), (d) and (e) only

175. If the anterior lobe of the pituitary gland gets completely damaged, then what will be the possible effect observed in males?

(1) Synthesis and secretion of androgen will increase

(2) Process of spermatogenesis will decline significantly

(3) Secretion of gonadotropin releasing hormone will decrease

(4) FSH concentration in blood will increase

176. A patient complains of excessive thirst and copious amount of urine. On examination of his urine sample, no glucose was detected but it was found to be very dilute (hypotonic). The patient is most likely suffering from

(1) Diabetes mellitus

(2) Diabetes insipidus

(3) Hypothyroidism

(4) Cretinism

177. **Assertion (A)** : Na^+ reabsorption from glomerular filtrate can occur in PCT, ascending limb of loop of Henle and collecting duct.

Reason (R) : Na^+ is always reabsorbed by passive process in PCT, ascending limb of loop of Henle and collecting duct.

In the light of above statements, choose the correct option

(1) Both (A) and (R) are true and (R) is the correct explanation of the (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of the (A)

(3) (A) is true statement but (R) is false

(4) Both (A) and (R) are false statements

178. Consider the given organisms.

(i) Marine fishes, (ii) Birds, (iii) Aquatic insects, (iv) Reptiles

Which of the following represents correct order of increasing toxicity of chief nitrogenous wastes secreted by respective organisms?

(1) (i) < (ii) = (iii) < (iv)

(2) (ii) = (iv) < (i) < (iii)

(3) (i) < (ii) = (iv) < (iii)

(4) (i) < (iii) = (iv) < (ii)

179. Out of 'X' number of total facial and the cranial bones in an adult human, 'Y' is the number of paired bones and 'Z' is number of unpaired bones. Choose the option which **correctly** represents number X, Y and Z.

	X	Y	Z
(1)	22	10	2
(2)	28	10	8
(3)	22	8	6
(4)	28	8	12

(1) (1)

(2) (2)

(3) (3)

(4) (4)

180. After kidney transplantation, which of the following hormones could be administered to the patient to prevent graft rejection?

(1) Mineralocorticoid

(2) Cortisol

(3) ADH

(4) ANF